

Schwarz Coupling for Solid Dynamics in Norma.jl: Transmission Conditions, Interface Energy, and Energy-Consistent Design

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Abstract

This note is the consolidated reference for the Schwarz coupling conditions implemented in Norma.jl and for the study of their energy behavior in dynamic solid mechanics. It records, in order: the Robin and impedance transmission conditions on shared (nonoverlap) interfaces and their reflection properties; the Dirichlet and impedance conditions on overlap boundaries, including the consistent partner traction that makes the impedance-overlap condition exact on conforming meshes (energy error 59% $\rightarrow$ 0.08% over 1 ms, with no adjustable parameters); a systematic transfer-fidelity study on nonconforming meshes showing that every increase in transfer fidelity worsens the interface energy exchange, because the residual error is a disagreement between the two grids' discrete dispersion relations rather than a transfer defect; a parametric energy study (151 configurations) showing that the interface work of the overlap coupling is sign-indefinite — neighboring mesh ratios swing from -53% to $+174\%$ to -66% drift — and that at a ten-millisecond horizon 74 of 108 nonconforming overlap runs diverge; the theoretical requirements for energy consistency (a single-valued interface traction acting through L^2 -adjoint transfer operators, which makes the interface power telescope to a sign-definite dissipation on the interface jump); and the resulting redesign. The adjoint-paired nonoverlap impedance coupling — one shared cross-mass matrix, force transfers adjoint to the partner's kinematic transfers, a harmonic-mean pair impedance, the dynamically consistent d'Alembert interface reaction, and an implicit-explicit (IMEX) treatment of the interface rows for explicit subdomains — conserves energy to 0.3% over 1 ms in all four integrator combinations on conforming and mildly nonconforming meshes, never injects energy at any mesh ratio, and holds the monodomain trajectory over ten thousand steps. Subcycling with genuine piecewise-linear time interpolation of partner data, the measured incompatibility of exact interface-work telescoping with a partitioned exchange, and the slot-keyed relaxation state required for controller stops larger than a subdomain time step complete the picture. Practical recommendations and open questions close the note.

This document consolidates and supersedes three earlier notes: “Impedance-Matching vs. Robin–Robin Coupling Conditions in Schwarz Methods,” “Variational Transfer for Impedance-Overlap Schwarz on Non-Conforming Meshes,” and “Interface Energy Loss of the Impedance-Overlap Schwarz Coupling on Nonconforming Meshes, and an Energy-Consistent Redesign” (June–July 2026).

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1 Setting

1.1 Decompositions and notation

Consider a solid body $\Omega \subset \mathbb{R}^3$ decomposed into two subdomains Ω_1 and Ω_2 that are integrated independently and coupled by a Schwarz iteration. Norma.jl supports two geometric configurations.

Nonoverlap. The subdomains partition Ω and share an interface $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ on which coupling is imposed. Let $\mathbf{n}_k$ be the outward unit normal to Ω_k on Γ (so $\mathbf{n}_1 = -\mathbf{n}_2$) and let $\mathbf{t}_k = \boldsymbol{\sigma}_k \mathbf{n}_k$ denote the traction on Ω_k 's side. Continuity of the coupled solution requires

$$\mathbf{u}_1 = \mathbf{u}_2 \quad (\text{kinematic continuity}), \quad (1)$$

$$\mathbf{t}_1 + \mathbf{t}_2 = \mathbf{0} \quad (\text{traction balance}). \quad (2)$$

Classical Dirichlet–Neumann Schwarz imposes (1) on one side and (2) on the other. The Robin and impedance variants studied here replace both with a linear combination of displacement, velocity, and traction (Sec. 2).

Overlap. The subdomains properly intersect: $\Omega_1 \cap \Omega_2 \neq \emptyset$. Each Ω_k has a portion of its boundary, $\Gamma_k \subset \Omega_j$ with $j \neq k$, lying strictly in the interior of the partner. That boundary is not shared with Ω_j , so no mutual traction is directly available there; on the other hand, the partner provides a well-defined interior state at every point of Γ_k . The coupling conditions available on an overlap boundary are those that can be built from that interior state (Sec. 3).

Throughout, $\mathbf{W}_k = \int_{\Gamma_k} \mathbf{N}_k \mathbf{N}_k^\top d\Gamma$ denotes the boundary mass matrix of side k , $\mathbf{P}$ a kinematic transfer operator mapping partner data to a side's interface nodes, a subscript p (in transmission conditions written per side) the partner field evaluated at this side's coupling boundary, and $\llbracket \cdot \rrbracket$ an interface jump.

1.2 Benchmark and energy metrics

All measurements in this note come from one benchmark: a bent 3D cantilever released from a parabolic initial displacement ($E = 6.895$ GPa, $\nu = 0.25$, $\rho = 2768$ kg/m³, bar length 0.254 m, $\Delta t = 10^{-6}$ s unless stated, horizons of 1 ms and 10 ms, Schwarz iterated to convergence at relative tolerance 10^{-8} at every step). Subdomain integrators are trapezoidal Newmark (implicit, I) and central difference (explicit, E), in the four pairings II, IE, EI, EE (first letter: clamped subdomain). Overlap decompositions split the bar lengthwise with overlap widths $\{0.0508, 0.0254, 0.0127\}$ m; the nonoverlap decomposition splits at $x = 0.127$ m. Nonconforming pairs fix the free subdomain's mesh at $h = 6.35$ mm and coarsen the clamped subdomain to $6.35/r$ mm, written as ratio 1: r ; the original three-point family used $\{1:1, 3:4, 1:2\}$ (i.e. $h = 8.47$ vs. 6.35 mm at 3:4).

Partition-of-unity (PU) energy. Summing the two subdomains' energies counts the overlap region twice and pollutes the diagnostic: a provably conservative run shows apparent $\pm 13\%$ excursions as the pulse transits the overlap. All energies quoted here count the overlap once,

through a partition of unity. The production implementation (`blended energy output: true; src/overlap_energy.jl`) uses a smooth Arlequin distance-ratio partition; the study's post-processor weights doubly-covered elements by 1/2 and reproduces Norma's unweighted output to at least four digits. Drift is reported as $E(t)/E(0) - 1$ in percent, with $E(0) \approx 1506$ J.

Staggered energy for central difference. Central difference advances the lumped-mass system with leapfrog structure, so the discrete energy it conserves exactly on linear problems is the staggered lumped-mass form

$$E_n = \frac{1}{2} \dot{\mathbf{u}}_{n-\frac{1}{2}}^T \mathbf{M}_L \dot{\mathbf{u}}_{n+\frac{1}{2}} + \text{SE}(\mathbf{u}_n) = \frac{1}{2} \dot{\mathbf{u}}_n^T \mathbf{M}_L \dot{\mathbf{u}}_n - \frac{\Delta t^2}{8} \mathbf{a}_n^T \mathbf{M}_L \mathbf{a}_n + \text{SE}(\mathbf{u}_n), \quad (3)$$

where $\mathbf{M}_L$ is the lumped mass and SE the stored energy. Evaluating instead the consistent-mass kinetic energy at integer steps misprices the short-wavelength content of the sharp release front twice over (the consistent and lumped Rayleigh quotients diverge at high wavenumber, and the integer-step velocity aliases the staggered content): a monodomain central-difference run of this benchmark reads -8.2% at 1 ms in the consistent-mass integer-step metric and $\pm 0.000\%$ in (3). Norma's energy output evaluates (3) for central-difference subdomains. Explicit-subdomain results below use this metric unless noted.

2 Transmission conditions on a shared interface

2.1 Robin coupling

On each side of the interface the user specifies a Robin parameter $\alpha > 0$ and imposes

$$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad k = 1, 2, \quad (4)$$

with the datum built from the partner's previous Schwarz iterate, $\mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}$, $j \neq k$. At the Schwarz fixed point $\mathbf{u}_k = \mathbf{u}_j$ and $\mathbf{t}_k = -\mathbf{t}_j$, so (4) reproduces (1) and (2) simultaneously.

Discretely, the self term contributes the boundary stiffness $\mathbf{K}_{\text{rs}} = \alpha \mathbf{W}$ to the tangent, and the partner datum enters the boundary force as $\mathbf{f}_{\Gamma,k} \leftarrow \mathbf{f}_{\Gamma,k} - \mathbf{t}_j|_{\Gamma} + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$. The parameter α has units of stiffness per unit area (N/m³) and interpolates between a Dirichlet constraint ($\alpha \rightarrow \infty$) and a Neumann condition ($\alpha \rightarrow 0$); the optimal value minimizes the Schwarz convergence rate and in practice the scaling $\alpha \sim E/h$ has been effective.

2.2 Impedance coupling

For a semi-infinite elastic bar with density ρ and P-wave modulus $M = \lambda + 2\mu$, a purely outgoing wave $u(x, t) = f(x - c_p t)$ with $c_p = \sqrt{M/\rho}$ satisfies $\dot{u} = -c_p u_x$, and with $\sigma = M u_x$,

$$\sigma + Z \dot{u} = 0, \quad Z = \rho c_p = \sqrt{\rho(\lambda + 2\mu)}, \quad (5)$$

the Lysmer–Kuhlemeyer nonreflecting boundary condition, with Z the characteristic acoustic impedance. The impedance Schwarz condition augments (4) with this absorbing term:

$$\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad \mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + Z \dot{\mathbf{u}}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}. \quad (6)$$

At the fixed point the impedance and displacement terms cancel and (6) again reproduces continuity and traction balance; the impedance is computed from material data with no user input, while α remains an optional user parameter.

Under Newmark integration with parameters β, γ the new-step velocity is linear in the new-step displacement with coefficient $c_v = \gamma/(\beta \Delta t)$, so the self term contributes the tangent $\mathbf{K}_{\text{is}} = (Z c_v + \alpha) \mathbf{W}$ and the self force $\mathbf{f}_{\text{is}} = \mathbf{W}(Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$; the partner datum enters the boundary force through the same surface operator and transfer as the Robin case with the additional velocity term. In the quasi-static limit Normajl sets $c_v = 0$ and the impedance condition collapses onto the Robin condition (4); note that a quasi-static impedance run with $\alpha = 0$ degenerates to a pure Neumann condition and does not couple the subdomains kinematically.

2.3 Reflection at the interface

The essential difference between the two conditions is their treatment of outgoing waves. Place the interface at $x = 0$ with Ω_k in $x < 0$ and consider a harmonic incident wave of amplitude F and a reflected wave of amplitude R . Imposing the Robin condition $\sigma + \alpha u = 0$ yields the reflection coefficient

$$R_{\text{RR}}(\omega) = \frac{i\omega Z - \alpha}{i\omega Z + \alpha}, \quad |R_{\text{RR}}(\omega)| \equiv 1 : \quad (7)$$

for any real $\alpha > 0$ the elastic interface fully reflects the incoming wave with a frequency-dependent phase shift, and no wave energy crosses the interface. Imposing the impedance condition $\sigma + Z\dot{u} + \alpha u = 0$ yields

$$R_{\text{imp}}(\omega) = -\frac{\alpha}{2i\omega Z + \alpha}, \quad |R_{\text{imp}}(\omega)| = \frac{\alpha}{\sqrt{\alpha^2 + 4\omega^2 Z^2}} < 1, \quad (8)$$

with $R_{\text{imp}} \equiv 0$ at $\alpha = 0$: outgoing waves are absorbed into the partner at all frequencies. This is the mechanism by which the impedance variant removes the reflection-driven energy growth that affects Robin coupling in dynamic problems, particularly when the two subdomains use different time integrators.

3 Transmission conditions on an overlap boundary

3.1 Dirichlet coupling

The classical overlap Schwarz condition replaces the free boundary of Ω_k on Γ_k with an imposed kinematic trace drawn from the partner,

$$\mathbf{u}_k = \mathbf{u}_j, \quad \dot{\mathbf{u}}_k = \dot{\mathbf{u}}_j, \quad \ddot{\mathbf{u}}_k = \ddot{\mathbf{u}}_j \quad \text{on } \Gamma_k, \quad (9)$$

constructed in one of two ways. *Pointwise interpolation*: for each overlap node, locate the partner element containing it in the reference configuration, evaluate the partner shape functions there, and assign the interpolated value; the corresponding degrees of freedom are then constrained. *Weak (L^2) projection*: precompute the projector $\mathbf{P}_{kj} = \mathbf{W}_k^{-1} \mathbf{L}_{kj}$, where $\mathbf{L}_{kj} = \int_{\Gamma_k} \mathbf{N}_k (\mathbf{N}_j^{\text{vol}})^\top d\Gamma$ is the cross-mass matrix coupling the partner's volume shape functions to the overlap boundary, and assign $\mathbf{u}_k = \mathbf{P}_{kj} \mathbf{u}_j$. Both variants are implemented by `SolidMechanicsOverlapSchwarzBoundaryCondition` (use `weak` flag). Because (9) constrains the overlap degrees of freedom, Dirichlet coupling adds no interface tangent term.

3.2 Impedance coupling

The impedance overlap condition leaves the overlap degrees of freedom free and couples Ω_k to the partner through a full zeroth-order optimized-Schwarz transmission condition,

$$\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z}(\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha(\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0} \quad \text{on } \Gamma_k, \quad (10)$$

where $\mathbf{t}_j = \boldsymbol{\sigma}_j \mathbf{n}_k$ is the partner's traction evaluated on Γ_k with Ω_k 's outward normal, and the impedance is the P/S-split tensor

$$\mathbf{Z} = Z_p \mathbf{n}_k \otimes \mathbf{n}_k + Z_s (\mathbf{I} - \mathbf{n}_k \otimes \mathbf{n}_k), \quad Z_p = \sqrt{\rho(\lambda + 2\mu)}, \quad Z_s = \sqrt{\rho\mu}, \quad (11)$$

with ρ, λ, μ taken from the partner's material (optimized-Schwarz cross-scaling: each side's optimal transmission operator approximates the neighbor's Dirichlet-to-Neumann map, so at a bimaterial interface each subdomain carries the other's characteristic impedances).

The partner traction is essential for consistency. The true monodomain solution has continuous kinematics and a generally nonzero traction $\mathbf{t}$ on the artificial surface Γ_k ; it satisfies (10) identically. Dropping $-\mathbf{t}_j$ (as an early implementation did) turns the condition into a relative dashpot, $\mathbf{t}_k = \mathbf{Z}(\dot{\mathbf{u}}_j - \dot{\mathbf{u}}_k) + \dots$, which the monodomain solution fails by exactly $\mathbf{t}$: the converged coupled solution then carries a spurious interface velocity jump $\approx \mathbf{t}/Z$ whose power drain $\sim |\mathbf{t}|^2/Z$ acts as permanent interface damping (56% energy loss over 1 ms on the conforming benchmark; Sec. 4).

Evaluating the partner traction. Because Γ_k is an interior surface of the partner mesh, the partner's assembled internal force cannot supply $\mathbf{t}_j$ there (interior rows carry the inertia residual, not $\boldsymbol{\sigma} \mathbf{n}$). Normajl evaluates $\mathbf{t}_j$ in one of two ways, selected by the **partner traction** option (default `auto`).

Consistent traction (node-aligned interfaces). When every node of Γ_k coincides with a partner node (detected automatically from the interpolation weights), the weak partner traction is extracted as the consistent nodal reaction: for each interface node i ,

$$[\mathbf{W} \mathbf{t}_j]_i = \int_{\Gamma_k} \mathbf{t}_j \varphi_i d\Gamma = - \sum_{e \in \mathcal{E}_i^{\text{ext}}} (\mathbf{f}_e^{\text{int}} + \mathbf{M}_e \ddot{\mathbf{u}}_e)_i, \quad (12)$$

where $\mathcal{E}_i^{\text{ext}}$ is the single layer of partner elements adjacent to Γ_k on its exterior side — precisely the elements Ω_k is missing from the monodomain assembly — and $\mathbf{M}_e$ is the partner's element mass (consistent or lumped, matching the partner's integrator). With this definition the monodomain solution satisfies Ω_k 's coupled discrete equations *identically*: the transmission condition transmits all discrete content, including the mesh-scale part that nodal stress recovery misrepresents. Implemented by `build_consistent_traction_patch!` and `consistent_partner_traction (schwarz.jl)`.

Recovered-stress traction (fallback). On nonconforming interfaces $\mathbf{t}_j$ is obtained by interpolating the partner's nodal-recovered stress with the same shape-function data used for the kinematic fields and contracting with Ω_k 's outward normals. Consistent (L^2 -projected, Cholesky-factorized) nodal recovery is force-enabled for any model carrying this condition; lumped recovery's $O(h)$ nodal-stress error feeds directly into the transmission condition and dominates the residual energy error (13.6% vs. 1.5% with consistent recovery; Sec. 4).

Discrete assembly. The self tangent consists of 3×3 nodal blocks $\mathbf{K}_{\text{is},ij} = W_{ij} (c_v \mathbf{Z}(\mathbf{n}_j) + \alpha \mathbf{I})$; the self force is $\mathbf{f}_{\text{is}} = \mathbf{W}(\mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$ with $\mathbf{Z}$ applied nodally before the surface weighting; and the partner datum enters as $\mathbf{f}_\Gamma \leftarrow \mathbf{f}_\Gamma + \mathbf{W} \mathbf{t}_j + \mathbf{W}(\mathbf{Z} \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$, with the kinematic fields transferred variationally by default (Sec. 8) or by pointwise interpolation as a legacy option. At the Schwarz fixed point all three jumps in (10) vanish for the monodomain solution, so on conforming meshes the converged coupled solution is the monodomain solution and the condition needs no parameter

tuning: $\mathbf{Z}$ is fixed by the partner material and α may be zero. A step-dependent multiplier on $\mathbf{Z}$ (`impedance_scale`) rescales the whole operator; with the traction term in place the conserving scale is the physical one (scale = 1), and the historical role of the scale as a per-problem tuning parameter — which compensated the missing-traction dissipation — is obsolete.

No separate Robin overlap variant. A Robin condition requires a partner traction for its datum; on an overlap boundary that traction is available only through the machinery above, and the resulting condition is exactly the $\mathbf{Z} = \mathbf{0}$ special case of (10). It is subsumed by the impedance overlap condition and not exposed separately.

3.3 Summary of the four conditions

Nonoverlap.

	Robin	Impedance
Boundary condition	$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$	$\mathbf{t}_k + \mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$
Self tangent	$\alpha \mathbf{W}$	$(\mathbf{Z} c_v + \alpha) \mathbf{W}$
Self internal force	$\alpha \mathbf{W} \mathbf{u}_k$	$\mathbf{W} (\mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$
Partner datum	$-\mathbf{t}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$	$-\mathbf{t}_j + \mathbf{Z} \mathbf{W} \mathbf{P} \dot{\mathbf{u}}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$
$\mathbf{Z}$ set by	(not used)	material (automatic)
α set by	user	user, optional
Quasi-static limit	same	collapses to Robin
Reflection $ R $	$\equiv 1$ (fully reflecting)	< 1 ; 0 at $\alpha = 0$

Overlap.

	Dirichlet	Impedance
Boundary condition	$\mathbf{u}_k = \mathbf{u}_j$ (pointwise or L^2)	$\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z} [\![\dot{\mathbf{u}}]\!] + \alpha [\![\mathbf{u}]\!] = \mathbf{0}$
Self tangent	(DOFs constrained)	$\mathbf{W}_{ij} (c_v \mathbf{Z}(\mathbf{n}_j) + \alpha \mathbf{I})$
Self internal force	(none)	$\mathbf{W} (\mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$
Partner datum	$\mathbf{u}_k \leftarrow \mathbf{P} \mathbf{u}_j$	$\mathbf{W} (\mathbf{t}_j + \mathbf{Z} \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$
Partner traction $\mathbf{t}_j$	(not used)	consistent (12) or recovered stress
$\mathbf{Z}$ set by	(not used)	partner material (automatic), P/S split
α set by	(not used)	user, optional

4 Conforming meshes: the partner-traction ladder

On the conforming, node-aligned overlap pair (II, PU metric, physical impedance, $\alpha = 0$), the energy error of the impedance overlap coupling over 1 ms decreases through four successive corrections, all parameter-free:

$$\underbrace{59\%}_{\text{missing } \mathbf{t}_j} \longrightarrow \underbrace{14\%}_{\text{+ traction term (lumped recovery)}} \longrightarrow \underbrace{1.5\%}_{\text{+ consistent recovery}} \longrightarrow \underbrace{0.08\%}_{\text{+ consistent traction}}$$

against the 0.1% reference of converged conforming Dirichlet coupling, which the final step matches.

Diagnosis of the recovered-stress residual. The 1.5–4% residual of the recovered-stress step was isolated with an interface-work instrumentation (`report_impedance_interface_work`, environment-gated, CSV output via `NORMA_IMPEDANCE_WORK_CSV`) that decomposes the interface power into partner-traction, dashpot, and Robin channels; the dashpot and Robin powers vanish identically for the exact solution, so their integrated work measures spurious dissipation directly. Three experiments discriminated the candidate mechanisms: (i) sweeping the Schwarz tolerance from 10^{-8} to 10^{-11} leaves the energy history bit-identical, ruling out iteration lag; (ii) the loss is invariant under $\Delta t \in \{2, 1, 0.5\} \times 10^{-6}$ s, ruling out the time discretization; (iii) the loss halves when h is halved ($3.9\% \rightarrow 1.9\%$): first order in mesh size. The measured velocity jump satisfies $\mathbf{Z}(\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) \approx \mathbf{t}_j - \mathbf{t}_k$ throughout — exactly what the converged transmission condition dictates — so the mechanism is: the recovered-stress traction error on mesh-scale field content is converted by the transmission condition into an interface velocity jump $\sim \delta t/Z$, which the dashpot dissipates at rate $|\delta t|^2/Z$ over the whole run. Replacing the recovered traction with the consistent traction (12) eliminates it: energy loss $3.93\% \rightarrow 0.076\%$ and integrated dashpot work $-42.6 \text{ J} \rightarrow +0.1 \text{ J}$ on the same benchmark. This also explains why Dirichlet overlap coupling is exact on conforming meshes — nodal injection transmits all discrete content with no recovery step.

P/S tensor split. The tensor impedance (11) is exact for body waves at normal incidence and is the correct operator for bimaterial interfaces. With the recovered-stress traction it appeared slightly detrimental on this homogeneous, flexure-dominated benchmark (conforming: 4.1% vs. 1.5% scalar); the diagnosis above resolves this: the tangential traction (recovery) error, which dominates in flexure, is dissipated at rate $|\delta t|^2/Z_s$, and $Z_s = Z_p/\sqrt{3}$ amplifies it. With the consistent traction the error term is gone and the tensor impedance conserves to 0.076%. Transmission conditions should be benchmarked on conforming meshes; nonconforming results can flatter a scheme through error cancellation (Sec. 5).

Impedance scale. With the traction term missing, a swept constant multiplier on Z interpolates between over-absorbing and unstable behavior, and the best scale on the nonconforming pair was 1.12 — a cancellation of the missing-traction dissipation against nonconforming transfer injection, hence mesh- and problem-specific. With the traction term in place the optimum is the physical scale 1.0; historical per-simulation tuning of the scale (and of large α values near 5×10^{11} – 10^{12}) optimized a double-counted diagnostic, not the physics.

5 Nonconforming overlap I: transfer fidelity

With the conforming case closed, the nonconforming overlap interface was studied on the 3:4 pair ($h = 8.47$ vs. 6.35 mm). The net drift looks small ($\pm 1\%$ at 1 ms), but the channel decomposition shows this is a cancellation of two error channels, each moving roughly half the system energy per millisecond:

1 ms, II, PU metric	conforming	nonconforming h	nonconforming $h/2$
dashpot (spurious) work	−42.6 J (2.8%)	−606.5 J (40%)	−598.5 J (40%)
traction-channel work	—	+677 J (45%)	+768 J (51%)
net PU-energy drift	1.5% $\rightarrow$ 0.08%	−0.75%	+1.5%
mean velocity jump (RMS)	8.9 m/s	40 m/s (growing 15 $\rightarrow$ 60)	27 m/s

Three facts drove the investigation: (i) the channels do not shrink under mesh refinement, so the pointwise-interpolation transfer feeds the condition an $O(1)$ error on mesh-scale content; (ii)

the net error is the difference of two large channels and flips sign between h and $h/2$ — it is not robust; (iii) the velocity jumps grow through the run: each interface pass leaves mesh-scale residue that reverberates between the grids.

5.1 The variational transfer and the fidelity ladder

Gander–Halpern–Nataf state three conditions for discrete energy convergence of waveform-relaxation coupling on nonmatching grids: (i) Robin (not Dirichlet) transmission; (ii) consistent boundary discretization of the transmission operator; (iii) an L^2 -contractive intergrid transfer. The impedance overlap condition satisfies (i) and (ii); pointwise interpolation violates (iii), while the variational projector $\mathbf{P}_{kj} = \mathbf{W}_k^{-1} \mathbf{L}_{kj}$ (Sec. 3.1) is an $L^2(\Gamma_k)$ -orthogonal projection, hence nonexpansive by construction, and transfers constants exactly for any quadrature rule (partition of unity gives $\mathbf{L}_{kj} \mathbf{1} = \mathbf{W}_k \mathbf{1}$).

A sequence of transfer improvements was implemented and measured on the instrumented 1 ms benchmark, each verified independently (the consistent constructions pass a uniform-stress patch test to 10^{-11} relative error):

- **Variational kinematic transfer** (`transfer: variational`): project $\dot{\mathbf{u}}_j$, $\mathbf{u}_j$, and the recovered stress variationally onto the receiving trace space instead of interpolating pointwise. On node-aligned interfaces the projector reduces to the identity, so the conforming limit is automatic.
- **Subdivided transfer quadrature** (`transfer quadrature subdivisions`): the integrand of $\mathbf{L}_{kj}$ is only piecewise smooth on Γ_k (partner elements change within a receiving facet), so an elevated sub-cell rule removes the consistency error of the default facet rule.
- **Consistent traction across the interface** (offset-surface construction): extract the consistent nodal reaction on the partner facet surface immediately exterior to Γ_k , where the one-sided partial assembly is exact, and transfer the weak traction to Γ_k ’s trace space with a surface-to-surface variational operator.
- **Single-operator characteristic transfer**: assemble the partner’s complete Robin datum $-\dot{\lambda} + \tilde{\mathbf{W}}(\mathbf{Z}\dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$ on the offset surface from the partner’s own nodal values and transfer it with one operator, so that any mesh-scale cancellation within the datum happens on the partner side, before transfer.

The measured ladder at resolution h (dashpot work and net drift at 1 ms):

1 ms, II, PU metric, h	dashpot work	net PU drift
pointwise + recovered stress	−607 J	−0.75%
variational + recovered stress	−726 J	+6.2% (growth)
variational, subdivided + recovered stress	−857 J	+2.4% (growth)
variational, subdivided + consistent traction	−1129 J	+9.3% (growth)
single-operator characteristic	−1084 J	+9.8% (growth)

Every fidelity increase makes the interface energy exchange larger. The variational transfer does restore an h -dependence that pointwise interpolation lacks (dashpot work 726 $\rightarrow$ 484 J under refinement, versus unchanged), but the error class does not change, and at production resolution the net worsens because projecting the kinematics decorrelates the kinematic-transfer error from the recovered-traction error, breaking the cancellation that kept the pointwise scheme’s net small. The

characteristic-transfer result refutes the hypothesis that the operator pairing between traction and velocity drives the residual. The conclusion is structural, not implementational: *on nonmatching grids, content that cannot be represented identically across the interface must be dissipated; smoothing the transferred data is beneficial, and nodal stress recovery had been acting as a load-bearing low-pass filter.*

5.2 Content-aware absorption: a decisive null result

To turn that accidental stabilization into a designed mechanism, an additional dissipative self term $\mathbf{W}\mathbf{Z}(\mathbf{I} - \Pi)\dot{\mathbf{u}}_k$ was implemented (**content aware absorption**), where Π is the $\mathbf{W}$ -orthogonal projection onto the span of the variational projector’s columns — the transferable subspace at the boundary. The operator is verified: it annihilates constants, vanishes identically on node-aligned interfaces and on the coarse side of the study pair, is nonzero only on the fine side ($\|\mathbf{F}\| = 0.44$), and is provably dissipative (asserted in test 105). **Its measured effect is null:** the absorber drains 4.6 J over 1 ms while the channels and the net remain identical to the baseline. Because the operator is verified, the null is informative: *the reverberating content lies inside the transferable subspace.* Both grids can represent the content that circulates; they disagree about it because their discrete dispersion relations differ. No transfer-side construction can remove a dispersion disagreement — which retroactively explains the entire fidelity ladder.

The closing diagnostic confirms the mechanism: rerunning the pointwise baseline with a dissipative integrator (Newmark $\gamma = 0.6$, $\beta = 0.3025$) collapses the dashpot work from -607 to -231 J and the velocity jumps *decay* ($11.8 \rightarrow 5.6$ m/s between run halves) instead of growing ($33 \rightarrow 47$): the reverberation loop is closed by subdomain-level damping, not by the interface. ($\gamma = 0.6$ is a blunt instrument — it also removed 87% of the physical energy on this broadband benchmark.)

5.3 HHT- α damping and its cost

HHT- α numerical dissipation was added to the Newmark integrator (**HHT alpha:** $\bar{\alpha} \in [0, 1/3]$; internal force assembled at the blended state $(1 - \bar{\alpha})u_{n+1} + \bar{\alpha}u_n$; $\gamma = 1/2 + \bar{\alpha}$ and $\beta = (1 + \bar{\alpha})^2/4$ derived; $\rho_\infty = (1 - \bar{\alpha})/(1 + \bar{\alpha})$) and swept on both mesh pairs at 1 ms:

$\bar{\alpha}$	0	0.02	0.05	0.1	0.2
nonconforming E/E_0	1.007	0.952	0.892	0.833	0.781
conforming E/E_0 (damping cost)	0.999	0.969	0.938	0.905	0.873
interface-specific extra loss	$\pm 1\%$	1.7%	4.9%	8.0%	10.5%
dashpot work [J]	-607	-595	-580	-561	-540
jump growth (halves, m/s)	$33 \rightarrow 47$	$32 \rightarrow 44$	$31 \rightarrow 41$	$30 \rightarrow 37$	$29 \rightarrow 32$

Mild HHT barely touches the interface channels even though $\bar{\alpha} = 0.1$ has the same $\rho_\infty = 0.818$ as the Newmark $\gamma = 0.6$ diagnostic: at $\Delta t = 10^{-6}$ s the mesh-frequency band sits at $\omega\Delta t \approx 0.3$ – 0.7 , which the time step resolves and HHT (by design) preserves, while broadband first-order Newmark- γ damping reaches it — and the physics with it (87% loss vs. 17% at the same ρ_∞). Damping converts the fragile $\pm 1\%$ net (sign-indefinite, growth-capable) into monotone, bounded decay at a cost proportional to how much of the problem’s energy lives in the reverberating band; on this benchmark that fraction is large by construction (the parabolic release has no scale separation), whereas production problems with smooth loading and resolution margin should obtain the robustness at small cost with $\bar{\alpha} \approx 0.02$ – 0.05 .

6 Nonconforming overlap II: the parametric energy study

6.1 Integrator–overlap–ratio sweep

Drift $E(1\text{ ms})/E(0) - 1$ in percent, pointwise transfer, across integrator pairs, overlap widths, and mesh ratios:

overlap [m]	ratio	II	IE	EI	EE
0.0508	1:1	−0.07	−8.9	−0.2	+8.7
	3:4	+1.1	−8.7	−0.9	−12.0
	1:2	+12.5	−26.5	−53.8	−57.7
0.0254	1:1	−0.07	+5.0	−0.4	+10.2
	3:4	−28.4	−6.0	−22.3	−13.9
	1:2	−17.3	−18.5	−58.9	−45.7
0.0127	1:1	−0.02	−15.7	−6.1	+3.6
	3:4	−44.5	−40.6	−41.1	−31.4
	1:2	−63.8	−64.3	−70.8	−61.9

Three regimes: (a) conforming implicit–implicit runs conserve to $\leq 0.1\%$ at every overlap width; (b) nonconforming runs lose tens of percent, worse with mesh ratio and with smaller overlap; (c) a minority of cases, including II at maximum overlap and 1:2, *gain* energy. Variational transfer improves the small-overlap losses substantially ($-44.5 \rightarrow -4.4\%$ at 0.0127/3:4) but regresses elsewhere ($+12.5 \rightarrow -50.7\%$ at 0.0508/1:2); it improved 17 of 24 nonconforming cases, reducing the mean |drift| from 33.4 to 25.4% — helpful, not structural.

6.2 Channel-resolved interface work

Integrated interface work over 1 ms (II runs, both subdomains):

case	ΔE blended	dashpot work	traction work	jump RMS
conforming (0.0254, 1:1)	−1 J	−1.8 J	+70 J	~ 1
nonconf. 0.0254, 3:4 (pointwise)	−428 J	−586 J	+92 J	~ 20
nonconf. 0.0127, 3:4 (pointwise)	−515 J	−877 J	+341 J	~ 16
nonconf. 0.0508, 1:2 (pointwise)	+188 J	+ 554 J	−211 J	~ 80
nonconf. 0.0127, 3:4 (variational)	−67 J	−864 J	+780 J	~ 26

Three facts follow. First, on conforming meshes the dashpot channel is negligible and the coupling is conservative: the velocity jump closes at Schwarz convergence. Second, on nonconforming meshes the dashpot channel carries the loss: the jump is an order of magnitude larger and never closes — the coarse trace space cannot represent the fine side’s boundary velocity, and the dashpot persistently absorbs the difference, exactly as an absorbing boundary absorbs outgoing content. Third, and decisively for the mechanism: in the 0.0508/1:2 case the “dashpot” does +554 J of *positive* work. A dissipator can never do positive work; this proves the discrete dashpot term is not a quadratic form. Because each subdomain interpolates partner fields with its own, unrelated operator, the two sides’ dashpot terms do not pair, and the total is sign-indefinite. The variational row explains why variational transfer helps only sometimes: its dashpot still drains −864 J, but the traction channel returns +780 J — the improvement is an accidental near-cancellation of two large opposing channels, not a removal of either defect.

Scaling both impedances by $s \in \{1, 0.5, 0.25\}$ on the 0.0127/3:4 II case reduces the loss monotonically (-44.5 , -40.5 , -29.2%) but sublinearly: the dissipated power is $Z[\dot{u}]^2$ and the jump itself grows as the absorber detunes, so the product falls more slowly than Z — the signature of an absorbing-boundary mechanism rather than of a fixed damping force.

6.3 Control experiment: Dirichlet matching

To separate “defect of the impedance condition” from “price of nonconforming overlap coupling,” the same nine mesh configurations were run with Dirichlet overlap coupling (II), in both transfer flavors:

overlap [m]	ratio	Dirichlet pointwise	Dirichlet weak
0.0508	1:1	+0.05%	—
	3:4	+5507%*	+2418%
	1:2	+2765%*	+5752%*
0.0254	1:1	−0.03%	—
	3:4	+423%	+749%
	1:2	+5628%*	+7765%*
0.0127	1:1	+0.01%	—
	3:4	+33%	+22%
	1:2	+3268%*	+262%

* run died (Newton failure) before 1 ms; drift at last stop.

Hard kinematic matching conserves essentially exactly on conforming meshes and is severely unstable on every nonconforming one — and the weak (contractive) transfer does *not* rescue it, consistent with the theory, since Dirichlet transmission violates the Robin requirement regardless of the transfer. Two consequences. First, the impedance dashpot’s dissipation is not merely a defect: it is what stabilizes a coupling that hard kinematic exchange destabilizes; its magnitude is the energy in the inter-grid disagreement it must continually absorb. Second, the Dirichlet instability grows with overlap width (opposite to the impedance losses), consistent with a feedback loop between the two interpolation surfaces through the overlap volume.

6.4 Mesh-ratio sweep: injection is generic

A finer-grained sweep (originating in a parametric study by our intern Lukas, then confirmed and extended) coarsens the clamped mesh through $r = 1.0, 0.9, \dots, 0.2$ at all three overlap widths and all four integrator pairs, with variational transfer throughout — 108 runs. The 1:1.0 column reproduces the conforming baselines of Sec. 6.1 to all printed digits, and 1:0.5 reproduces the earlier variational 1:2 results exactly.

ratio	overlap 0.0508 m				overlap 0.0254 m				overlap 0.0127 m			
	II	IE	EI	EE	II	IE	EI	EE	II	IE	EI	EE
1:1.0	-0.1	-8.9	-0.2	+8.7	-0.1	+5.0	-0.4	+10.2	-0.0	-15.7	-6.1	+3.6
1:0.9	+7.6	-11.7	-5.5	+8.5	-4.4	-8.1	+4.4	+9.1	+0.1	-14.8	-1.0	+4.3
1:0.8	+29.8	-15.3	+5.5	-9.5	+20.4	+45.1	-2.4	-1.9	-3.1	-20.3	-5.4	+4.7
1:0.7	+19.2	-12.8	+2.7	-11.6	+11.8	-18.6	+0.7	-5.4	+11.1	+7.3	-9.8	-17.9
1:0.6	-61.1	-54.0	-57.7	-40.4	+2.1	-18.6	-63.9	-53.4	-4.5	-34.2	-57.1	-52.7
1:0.5	-50.6	-32.3	-36.4	-56.8	-11.0	-17.6	-55.7	-54.3	-29.2	-54.0	-56.8	-52.0
1:0.4	-12.4	+3.9	-43.1	-53.1	-52.7	-50.2	-57.0	-63.8	-36.6	-52.4	-78.2	-76.2
1:0.3	-65.0	-44.5	-78.3	-57.4	+173.7	+274.7	-81.2	-78.8	-30.5	-58.6	-79.5	-79.4
1:0.2	-79.1	-76.8	-80.8	-78.9	-66.5	-66.2	-86.3	-87.4	-22.2	-45.8	-91.2	-90.3

Three observations sharpen the diagnosis.

Injection at mild coarsening is systematic, not exceptional. The implicit-clamped pairs inject at nearly every overlap for ratios 1:0.9–1:0.7: II +29.8%, IE +45.1%, and seven further cells above +7% — with the contractive variational transfer already in place. The contraction property bounds the transfer but not the sign of the dashpot work; no per-side transfer improvement yields sign-definiteness on the two-surface overlap coupling.

An injection resonance. At 0.0254/1:0.3 the coupling drives the beam to +174% (II) and +275% (IE) of $E(0)$, growing smoothly for ~ 0.7 ms and saturating near $3\text{--}4 \times E(0)$ — steady interface pumping, not a numerical divergence. At this ratio the clamped mesh has degenerated to a single-element cross-section, but degeneracy alone does not decide the outcome: the same clamped mesh at quarter overlap dissipates -30.5%, and the explicit-clamped pairs on the identical configuration dissipate -81%. Whether the sign-indefinite interface work injects or absorbs is a property of the mesh pair and the overlap geometry together.

The drift is strongly non-monotone in the ratio. At 0.0254 overlap, II moves $-52.7 \rightarrow +173.7 \rightarrow -66.5\%$ across the three adjacent ratios $1:0.4 \rightarrow 1:0.3 \rightarrow 1:0.2$. There is no usable regime map of the form “mild contrast implies small drift”: both the sign and the magnitude of the energy defect are extremely sensitive to the discrete mesh pair.

This closes the question of whether the nonconforming overlap coupling can be trusted with a drift budget: it cannot. The interface work is sign-indefinite in practice as well as in principle, and both signs are realized at large amplitude across neighboring points of a smooth parameter family.

7 Theoretical requirements for energy consistency

Four independent lines of analysis agree on the structure of the problem and of its solution.

7.1 The interface-work pairing identity

For Newmark-family subdomains coupled by interface tractions, the discrete energy balance over a step contains the interface term (Gravouil–Combescure 2001, Eq. (34); Karimi–Nakshatrala 2014, Eq. (84))

$$E_{\text{int}} = \sum_{\text{sides } k} \frac{1}{\Delta t_k} [\dot{\mathbf{u}}_k]^T \mathbf{C}_k^T [\Lambda_k], \quad (13)$$

the work of the transmitted tractions through the kinematic jump, each side on its own grid. It vanishes if and only if the traction is a single-valued field acting equal-and-opposite on the two sides

through *adjoint* trace operators, with velocity continuity enforced at instants shared by both time grids. Gravouil–Combescure prove their subcycled coupling makes $E_{\text{int}} \leq 0$ (dissipative, never a source); Prakash–Hjelmstad restructure the intermediate tractions so it telescopes to exactly zero. Neither treats spatially nonconforming interfaces; both warn that breaking the pairing leaves E_{int} of indefinite sign — precisely the +554 J observation of Sec. 6.2.

7.2 Representability: the converged nonconforming solution

Gander–Halpern–Nataf (SINUM 2003) analyze discrete Schwarz waveform relaxation with impedance transmission on nonmatching grids. Their key statements: the choice $Z = \rho c$ of the partner side is the exact transparent operator in one dimension and optimal-in-class generally, so the impedance choice is not the problem; their nonmatching-grid convergence proof requires the intergrid transfer to be an L^2 contraction (a projection qualifies; pointwise interpolation does not); and, for nonconforming grids, the coupled system *defines* the solution when converged — there is no monodomain scheme it equals. Continuity at the converged fixed point holds only through the projection: the component of the fine trace orthogonal to the coarse trace space is invisible to the partner and never matched. The impedance dashpot then acts on that persistent component as a genuine absorbing boundary — sign-definite dissipation that is the price of stability, not an iteration defect. The cure they and Achdou–Japhet–Maday–Nataf (Numer. Math. 2002) point to is not a better Z but a *common trace space* for the datum entering the dashpot.

7.3 The telescoping condition and the adjoint requirement

Energy-stable discontinuous-Galerkin couplings (Appelö–Wang 2019; Duru et al. 2021; Liu et al. 2023) make the interface energy identity exact by construction: a single Riemann state is computed on a common description of the interface (the mesh intersection) and enters both sides’ weak forms; the interface power then telescopes to

$$P = - \int_{\Gamma} [\![\dot{\mathbf{u}}]\!]^T \mathbf{Z} [\![\dot{\mathbf{u}}]\!] dS \leq 0 \quad (\text{or exactly 0 for the central variant}), \quad (14)$$

with the dissipation confined to the jump and vanishing under refinement and at convergence. The precise discrete requirement: if side k receives partner kinematics through $\Pi_{j \rightarrow k}$, the transfers must be L^2 -adjoints through the variational cross-mass matrix,

$$\mathbf{M}_1 \Pi_{2 \rightarrow 1} = (\mathbf{M}_2 \Pi_{1 \rightarrow 2})^T = \mathbf{B}, \quad B_{mn} = \int_{\Gamma} \varphi_m^1 \varphi_n^2 dS, \quad (15)$$

with $\mathbf{M}_k$ the side- k boundary mass matrices and the integrals evaluated on the common refinement. Two independent pointwise interpolations violate (15), the cross terms do not cancel, and the interface term has no sign. Liu et al. additionally warn that nonconforming cross-interpolation matrices are severely ill-conditioned and that interface unknowns may need implicit treatment even between explicit interiors (their IMEX-Newmark) — directly relevant to the conforming IE/EE growth (+5 to +10% in Sec. 6.1), where the explicitly evaluated dashpot at lagged time levels injects energy that central difference ($\gamma = \frac{1}{2}$, zero algorithmic damping) never removes.

7.4 Synthesis: why each observed regime occurs

Conforming, implicit. The transfer is the identity, the adjoint condition (15) holds trivially, the jump closes at Schwarz convergence, the dashpot does no work: conserved.

Nonconforming, dissipative majority. The jump cannot close (representability); the dashpot absorbs the unrepresentable content persistently. The loss grows with mesh ratio and shrinks with overlap width (a wider overlap buffers the boundary from fine content and gives the coarse subdomain more elements to express the transferred field).

Nonconforming, growth cases. Non-adjoint pairing makes the dashpot term sign-indefinite; wherever the positive branch dominates, the coupling injects instead of absorbing. The ratio sweep shows this is generic and that the sign flips between adjacent mesh ratios: structural, not stochastic.

Conforming, explicit free subdomain. A time-level, not spatial, pairing failure: the dashpot force is evaluated explicitly at the lagged partner velocity, and central difference has no algorithmic damping to absorb the error.

Variational improvement. The variational projection is a contraction and the characteristic transfer restores much of the traction consistency, but the dashpot still sees the full jump; the net outcome rides on near-cancellation of two large channels.

8 Redesign within the overlap family

Four redesign phases were formulated from the theory; their outcomes:

Phase 1 — variational transfer by default (implemented, reframed). The full adjoint pairing of (15) presumes a *shared* interface on which the two sides’ works telescope. The overlap variant has none: data are exchanged on two distinct surfaces Γ_1, Γ_2 , so no pairing of their works exists, and energy consistency can come only from agreement of the two solutions in the overlap — exactly what dispersion mismatch precludes at fixed h (Sec. 5.2). The implementable core of Phase 1 is the contraction requirement: **transfer: variational** is the default for the impedance overlap condition (pointwise remains an explicit legacy option; the two coincide on node-aligned interfaces). Empirically this converts the worst sign-indefinite case (+554 J of positive dashpot work) into sign-correct controlled dissipation and improves 17 of 24 nonconforming cases, but it cannot eliminate the dispersion-jump dissipation, and isolated growth cells persist.

Phase 2 — dashpot on the representable jump (implemented; falsified). Replace $\mathbf{Z}[\hat{\mathbf{u}}]$ by $\mathbf{Z} \hat{\Pi}[\hat{\mathbf{u}}]$, where $\hat{\Pi}$ is the W -orthogonal projection onto the transferable span (**representable dashpot: true**; a strict no-op on node-aligned interfaces). The premise was that the persistent jump is trace content the coarse space cannot represent, so that filtering it would let it reflect (conserving) instead of being absorbed. Measurement falsifies the premise: on the 0.0127/3:4 case the fine side’s $\hat{\Pi}$ is a genuine projection ($\|\hat{\Pi} - I\|/\sqrt{n} = 0.48$) yet the dashpot work is unchanged ($-1066 \rightarrow -1065$ J) — the jump lies *inside* the representable subspace. The persistent jump is a solution-content defect, not a spatial-representability defect: coarse-grid dispersion phase-lags the transiting pulse, so at Schwarz convergence the two discrete solutions disagree by a smooth, in-band field that no static spatial filter can remove. The option is retained (default off) as a diagnostic of where the jump lives.

Phases 3 and 4 — harmonic-mean impedance; implicit interface for explicit subdomains. Both belong structurally to the nonoverlap variant, where they were implemented as part of the adjoint pairing (Sec. 9): the pair impedance $2Z_1 Z_2 / (Z_1 + Z_2)$ per wave family (the Riemann-solver weighting of Duru et al. and Liu et al.), and the IMEX interface treatment (Sec. 9.3).

The channel-resolved interface-work instrumentation remains the regression metric: the dashpot work must be ≈ 0 on conforming meshes at Schwarz tolerance, and the study’s drift tables are the

acceptance benchmark for any future change to the transmission machinery. A universal dashpot sign test is *not* attainable in the overlap variant, for the structural reason stated under Phase 1.

9 The adjoint-paired nonoverlap coupling

The theory identifies the nonoverlap impedance variant — one shared interface Γ , transmission $\mathbf{t} + Z\dot{\mathbf{u}} + \alpha W\mathbf{u} = \mathbf{g}$ — as the member of the family on which the adjoint condition (15) is actually available. It has been implemented and measured on the cantilever benchmark, split at $x = 0.127$ m, with the mesh-ratio family of Sec. 6.4.

9.1 Baseline: the legacy coupling

The pre-existing nonoverlap impedance coupling (per-side transfer operators, per-side impedance and Robin parameter) loses 7–18% at mild mesh contrast — *including on conforming meshes* (−9.6% at 1:1) — and injects strongly at strong contrast (+223% EI and +363% EE at 1:0.3; +77% II at 1:0.2). Probes isolate the conforming loss: it is unchanged under $\alpha \rightarrow 0$, under equalized α , and under $\Delta t \rightarrow \Delta t/2$, with the Schwarz iteration converged to 10^{-8} . It is therefore a *consistency* defect of the fixed point itself: the coupling transfers the partner’s static reaction $-\mathbf{f}^{\text{int}}$, but in dynamics the two sides’ static reactions are not equal and opposite — they differ by the interface nodes’ inertia, each side carrying its own tributary mass. The coupled fixed point then differs from the monodomain discretization by $O(m_\Gamma a)$, measured as a steady, Δt -independent interface drain.

9.2 The paired coupling

Enabling adjoint pairing: `true` (the default for Schwarz impedance nonoverlap; adjoint pairing: `false` restores the legacy transfer) changes three things:

1. the exchanged traction becomes the dynamically consistent d’Alembert reaction $\mathbf{M}\mathbf{a} + \mathbf{f}^{\text{int}} - \mathbf{f}^{\text{body}}$, so the two interface rows sum exactly to the monodomain row and the monodomain trajectory is the fixed point;
2. both sides derive their transfer operators from *one* shared cross-mass matrix B (integrated over the finer side’s facets), with $\Pi_1 = W_1^{-1}B$, $\Pi_2 = W_2^{-1}B^T$, and each side’s force transfer the adjoint of the partner’s kinematic transfer ($N_1 = \Pi_2^T$, $N_2 = \Pi_1^T$), which is exactly (15);
3. the pair shares one impedance — the harmonic mean $2Z_1Z_2/(Z_1+Z_2)$ of the per-side material impedances $Z_k = \rho_k c_{p,k}$ — and one Robin parameter α .

On parameter selection: the impedance is *estimated* from material data, the Robin parameter is *chosen* by the user (`robin parameter`, default 0), and no estimate of α is attempted because none is needed for consistency. At the Schwarz fixed point the two Robin conditions add to traction balance and subtract to $Z\delta + \alpha W\delta = 0$ for the interface jump δ ; with $\delta(0) = 0$ the converged solution is independent of $Z > 0$ and $\alpha \geq 0$ — both act only on the convergence rate of the iteration and on the interface energy bookkeeping en route. What consistency does require is that α be one number per interface: the Robin spring stores and returns energy conservatively only when both sides’ interface forces pair through the same coefficient. Mismatched values under pairing are therefore a hard input error — Norma aborts, naming both side sets and values — since silently averaging them would apply a condition neither side stated.

Drift at 1 ms, implicit–implicit:

ratio	legacy II	paired II
1:1.0	−9.61	− 0.03
1:0.9	−6.84	− 0.03
1:0.8	−7.96	−3.59
1:0.7	−2.06	−6.55
1:0.6	−9.39	−9.71
1:0.5	−13.25	−13.32
1:0.4	−45.78	−15.13
1:0.3	+18.95	−15.08
1:0.2	+77.49	−13.26

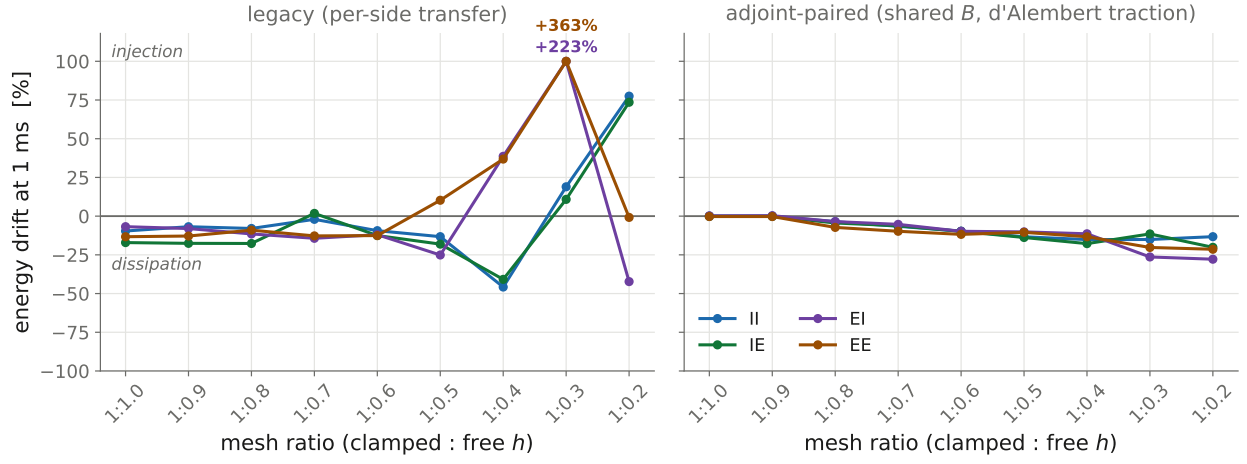


Figure 1: Nonoverlap impedance Schwarz, energy drift at 1 ms across the mesh-ratio family, all four integrator combinations. Left: legacy per-side transfer (consistent-mass integer-step metric) — sign-indefinite, with injection up to +363% (EE) and +223% (EI) at 1:0.3. Right: the adjoint-paired coupling (staggered lumped-mass metric of Eq. (3) for explicit members) — monotone, sign-definite trace-jump dissipation at every ratio, and conservation to $\leq 0.3\%$ on conforming and 1:0.9 meshes.

Three properties, in order of importance. First, *no injection at any ratio*: the interface work is sign-definite, as the telescoping identity (14) guarantees. Second, near-conservation through mild contrast (-0.03% at 1:0.9, where every other variant tested loses or gains several percent). Third, the residual dissipation at strong contrast is the genuine trace-space jump being absorbed at rate $Z \int_{\Gamma} \llbracket \dot{\mathbf{u}} \rrbracket^2$: at the exactly nested ratio 1:0.5 (where the legacy transfer is coincidentally adjoint and the quadrature is exact) paired and legacy agree to 0.1%, so the remaining loss is not a transfer artifact but the representability price of Sec. 7.2 — now paid in a controlled, monotone, sign-definite form.

9.3 Explicit subdomains: IMEX interface treatment

With an explicit (central-difference) subdomain on either side, the paired coupling is strongly unstable ($+10^3$ to $+10^5\%$) if the interface terms are evaluated at lagged time levels: the consistent traction couples the partner’s *acceleration* into the interface force, an algebraic loop that the implicit pairs resolve within the Schwarz iteration. The cure is the IMEX structure of Liu et al.: because

the dashpot force is linear in the unknown acceleration, the explicit update is closed implicitly on the interface rows only,

$$(\mathbf{M} + \gamma \Delta t \mathbf{Z} \mathbf{W})|_{\Gamma} \mathbf{a}_{n+1} = \mathbf{M} \mathbf{a}^{\text{expl}}|_{\Gamma}, \quad (16)$$

a small symmetric-positive-definite solve per interface (reused for the three components) while the interior stays matrix-free explicit (`imex_interface_acceleration!`); the dashpot then acts on the end-of-step velocity and the Schwarz iteration synchronizes both sides at t_{n+1} exactly as in the implicit–implicit case. Drift at 1 ms, all combinations, staggered metric (3) for explicit members:

ratio	II	IE	EI	EE
1:1.0	−0.03	−0.11	+0.20	−0.27
1:0.9	−0.03	−0.10	+0.29	−0.27
1:0.8	−3.6	−4.4	−3.5	−7.3
1:0.7	−6.6	−6.2	−5.3	−9.8
1:0.6	−9.7	−9.8	−9.8	−11.8
1:0.5	−13.3	−13.8	−10.2	−10.5
1:0.4	−15.1	−17.8	−11.4	−13.3
1:0.3	−15.1	−11.5	−26.4	−20.2
1:0.2	−13.3	−20.1	−27.8	−21.4

On conforming and mildly nonconforming meshes all four integrator combinations conserve to $\leq 0.3\%$; the nonconforming columns carry the same monotone, sign-definite trace-jump dissipation in every combination. (The small positive EI entries, $+0.2$ – 0.3% , are bounded transient storage in the conservative interface spring, not secular growth.) In the consistent-mass integer-step metric the explicit columns instead read -4 to -37% ; the difference is the metric artifact of Sec. 1.2, not the coupling — against the monodomain explicit reference the conforming paired EE trajectory is exact to 0.35% *pointwise over the whole millisecond*: the coupled trajectory is the monodomain trajectory, as the fixed-point argument predicts.

10 Subcycling: different time steps per subdomain

Different time steps per subdomain require no new exchange machinery: each subdomain records its per-substep trajectory (kinematics and internal force) during every Schwarz iteration, the coupling evaluates the partner’s state in time at each of its own substeps, and the paired d’Alembert traction follows. (Enabling this exposed a time-step erosion defect in the subcyclers — the stop-aligned remainder step persisted as the nominal step and the Schwarz re-integration amplified the floating-point remainder geometrically — fixed independently of the pairing.)

10.1 Genuine interpolation

The exchange was intended to interpolate the partner piecewise-linearly in time — the interpolated-continuity structure under which Gravouil–Combescuré prove the subcycled interface work dissipative ($E_{\text{int}} \leq 0$, $O(\Delta t)$) — but two infrastructure defects meant that is not what actually ran. First, the history interpolation selected the wrong segment (an off-by-one that returned the end-of-history value for any query in the last segment and extrapolated backward elsewhere). Second, the per-stop history was anchored at the first substep, not the stop start, so a non-subcycling partner’s history was a single end-of-stop entry. Net effect: both directions of the exchange end-sampled the partner’s stop-end state — constant, not interpolated, data. With both defects fixed the exchange

is genuinely piecewise-linear in both directions. Drift at 1 ms, free side subcycled 4:1 ($\Delta t = 0.25 \mu\text{s}$ inside a $1 \mu\text{s}$ stop; consistent-mass integer-step metric for comparability across the development history):

case	end-sampled (defective)	interpolated (fixed)
conforming II	−2.8	−3.6
conforming IE	−9.6	−10.1
conforming EE	−10.6	−9.6
1:0.5 II	−20.6	− 14.1
1:0.5 IE	−24.4	− 21.3

(In the staggered metric the interpolated exchange reads conforming II −3.6, IE −1.4, EE −1.2%: the explicit subcycled overhead is about one percent on conforming meshes.) Genuine interpolation wins decisively at mesh contrast, where end-sampling aliases the fine side’s intra-window motion into the exchanged traction. It loses slightly on the conforming implicit cases because the end-sampled datum, being the stop-end value applied throughout the window, *leads* the true trajectory, and that anticipatory bias is a mild energy-injection channel that happened to offset part of the interpolation dissipation. A cancellation between two discretization errors is not a conservation property, and it collapses exactly where the stakes are high; the interpolated exchange is the behavior of the paired coupling.

10.2 Exact telescoping is incompatible with a partitioned exchange

Prakash–Hjelmstad telescope the subcycled interface work to exactly zero by giving the interface traction a linear-in-time representation whose step values satisfy $\frac{1}{2}(\hat{\lambda}_n + \hat{\lambda}_{n+1}) = \bar{\lambda}_{n \rightarrow n+1}$ (the partner’s window average), so the trapezoidal work sums of the two sides cancel window by window. Three Schwarz-compatible constructions of this datum were implemented and measured:

Window average. Handing the coarse side the partner’s trapezoidal window average is work-conjugate to the fine side’s interpolation — but an average estimates the window midpoint, and the half-window lag of the interface force behind the interface motion is artificial damping: uniformly worse than plain interpolation (−3.6 $\rightarrow$ −4.2, −10.1 $\rightarrow$ −11.1, −9.6 $\rightarrow$ −12.3, −14.1 $\rightarrow$ −15.5, −21.3 $\rightarrow$ −23.0).

Exact telescoping recursion. The lag-free datum $\hat{\lambda}_{n+1} = 2\bar{\lambda} - \hat{\lambda}_n$ satisfies the telescoping identity exactly — and diverges: the factor of two on the partner-state channel doubles the Schwarz iteration’s contraction factor, and the coupled iteration diverges within tens of stops.

Least-squares endpoint fit. The endpoint value of the least-squares linear fit of the partner trajectory over the window is lag-free, filters intra-window content instead of aliasing it, and telescopes for window-linear tractions. It is stable over tens of stops and passes the short regression — and over hundreds of stops it injects energy catastrophically ($+10^3$ – $10^5\%$ before aborting): a zero-lag filtered estimate is an extrapolation, and its frequency response exceeds unity off the linear band.

The pattern is the classical stability principle of partitioned coupling: only *causal* transfer operators (lag ≥ 0) are robust in a fixed-point exchange, and every zero-lag work-conjugate datum is anticipatory. Prakash–Hjelmstad’s telescoping is consistent with this — their method solves the coupled interface problem *directly* (a FETI-style interface system per large step), which is

precisely what a partitioned Schwarz sweep does not do. Exact subcycled interface-work telescoping therefore requires a direct interface solve, not a smarter exchange; within the Schwarz family, genuinely interpolated Gravouil–Combescure transfer is the measured stable optimum, and its $O(\Delta t)$ dissipation (-3.6% at conforming II vs. -0.03% same-step) is the price of subcycling. (The window-average and endpoint-fit utilities remain in `interpolation.jl` as tested building blocks.)

11 Long-horizon behavior

Every conclusion above rests on 1 ms runs — about two transits of the release front. To test which behaviors are steady and which are transients, all 151 study configurations were rerun to 10 ms (10^4 stops): the full overlap sweep, the nonoverlap paired sweep and its subcycled set, and fresh monodomain references. Both monodomain references conserve the staggered energy (3) to $\pm 0.000\%$ over the full 10 ms, and every surviving run reproduces its recorded 1 ms drift at its $t = 1$ ms row.

11.1 Overlap: the 1 ms drifts were transients of instabilities

The overlap family is long-horizon unstable almost everywhere. Of the 108 overlap runs, 74 died of element inversion after exponential energy growth (typically $+3,000$ to $+10,000\%$ of E_0 at death, between 2 and 9 ms), and most of the remainder collapsed toward zero energy. The sign-indefinite drifts of the 1 ms sweep were early sections of these trajectories: configurations that read as dissipative at 1 ms (e.g. maximum overlap 1:0.6 II, -61%) reversed and grew to $+5,432\%$ at death. Mesh conformity does not save the mixed-integrator combinations: the conforming half-overlap IE run read $+13\%$ at 1 ms, grew cleanly exponentially, and died at 6.25 ms; conforming EI behaves alike at all three widths. The only healthy overlap configurations at 10 ms are the conforming II runs: $+0.64$, $+0.61$, $+0.16\%$ at the three overlap widths (Fig. 2). The overlap coupling is therefore a short-horizon method on nonconforming meshes, and for non-II combinations even on conforming ones.

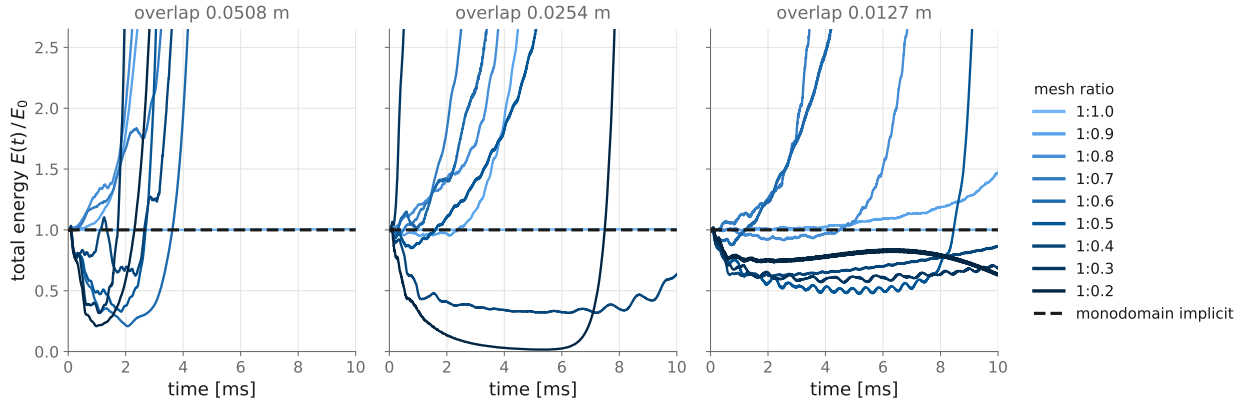


Figure 2: Overlap impedance Schwarz (variational transfer), implicit–implicit, 10 ms horizon. Nearly every nonconforming configuration either grows exponentially until element inversion ends the run (curves leaving the top of the axes) or collapses toward zero energy; only the conforming (1:1.0) runs track the monodomain reference. The mixed and all-explicit combinations (not shown) are worse: they lose even the conforming cases.

11.2 Nonoverlap pairing: no injection, but compounding dissipation and two new failure modes

The paired nonoverlap coupling never injects at any ratio in any combination, at 10 ms as at 1 ms. Its monotone trace-jump dissipation, however, compounds. Drift at 10 ms ($\dagger t$ marks a run dead by element inversion at t ms; EE entries marked * ended in the noise-saturated state discussed below):

ratio	II	IE	EI	EE
1:1.0	+0.00	−0.9	−0.2	*
1:0.9	+0.00	−0.5	+0.1	*
1:0.8	$\dagger$ 5.2	$\dagger$ 3.9	$\dagger$ 4.0	$\dagger$ 2.6
1:0.7	$\dagger$ 4.9	$\dagger$ 3.7	$\dagger$ 3.9	$\dagger$ 2.8
1:0.6	−47.3	−56.4	−42.9	*
1:0.5	−56.8	−72.0	−40.8	*
1:0.4	−54.9	−73.9	−40.4	*
1:0.3	−65.1	−66.0	−81.1	*
1:0.2	−53.3	−73.1	−84.1	*

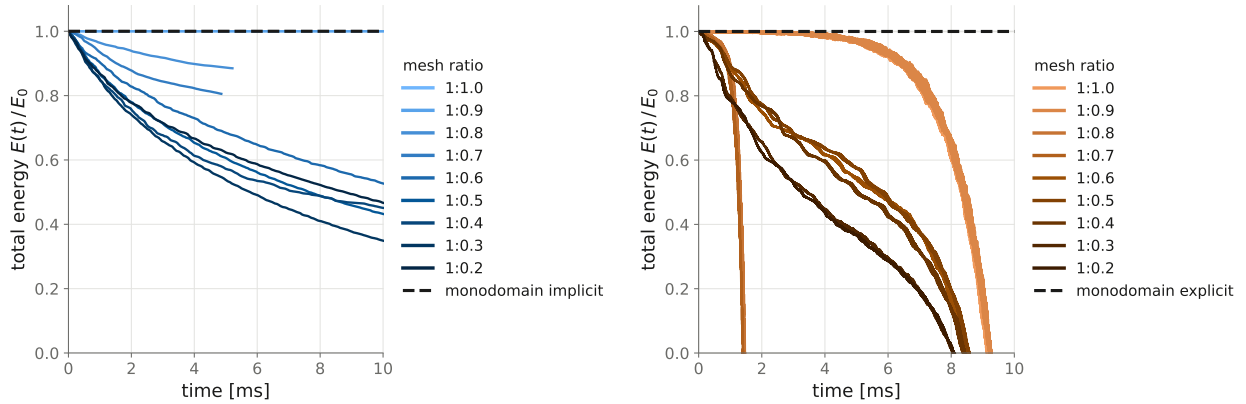


Figure 3: Adjoint-paired nonoverlap impedance Schwarz, 10 ms horizon, staggered energy metric. Left: implicit–implicit — no injection anywhere; conforming and 1:0.9 hold the monodomain reference, strong contrast dissipates monotonically, and the non-nested ratios 1:0.7–1:0.8 end early at ≈ 5 ms (interface element inversion). Right: explicit–explicit — the staggered kinetic term goes secularly negative at every ratio as undamped grid-frequency content accumulates at CFL = 1; the 1:0.7–1:0.8 runs die of central-difference instability at 2.6–2.8 ms.

Three observations (Fig. 3). First, the conforming and 1:0.9 columns with an implicit member are excellent — the coupling holds 0.00 to −0.9% over ten thousand stops. Second, at strong contrast ($\leq 1:0.6$) the sign-definite dissipation that read as 10–28% at 1 ms reaches 40–85% by 10 ms: bounded and ordered, but far too large for wave-transit accuracy. Absorption of unrepresentable content is a rate, not a one-time toll. Third, the mildly nonconforming, *non-nested* ratios 1:0.7 and 1:0.8 form a failure window invisible at 1 ms: the paired transfer accumulates interface-local high-frequency displacement content, and every integrator combination eventually dies there — explicit members at 2.6–4 ms by central-difference instability, implicit pairs at ≈ 5 ms when the first interface element inverts while the global energy is still smooth (−19% at death). This is not a relaxation artifact: reruns with fixed $\theta = 0.5$ die identically (trajectories matching to three digits),

so the noise accumulation belongs to the transfer operators themselves. The nested ratios (1:0.5 exactly, and the strong ratios whose transfer filters aggressively) do not exhibit it.

11.3 Explicit–explicit saturation, and subcycling as a remedy

The EE column degrades at every ratio by 10 ms, conforming included, ending with the staggered metric near $-2 E_0$. The end state is not a divergence: the stored energy stays bounded and physical throughout (the conforming run holds $E/E_0 \geq 0.97$ to 5 ms, then slides), but the staggered kinetic term goes steadily negative — the signature of sawtooth content at $\omega\Delta t \approx 2$ accumulating in velocity and acceleration. Central difference at CFL = 1 has zero numerical dissipation at the stability edge, so whatever grid-frequency content the interface exchange seeds is never removed; with no implicit member in the pair, it accumulates without bound in amplitude-neutral modes. The measured remedy is time-step margin: the subcycled conforming EE run (free side at $\Delta t/4$, CFL = 0.25) ends at -10.8% with no saturation. Subcycling cuts the other way for II: the same-step conforming II run conserves to $+0.00\%$ while the subcycled one loses -20.2% (at 1:0.5, -61.6%) — the Gravouil–Combesure interpolation dissipation of Sec. 10.1, harmless per stop, compounds over 10^4 stops. At long horizons the multi-time-step exchange is thus a trade: it stabilizes explicit members (CFL margin) and taxes implicit ones (interpolation dissipation). Figure 4 shows both sides of the trade.

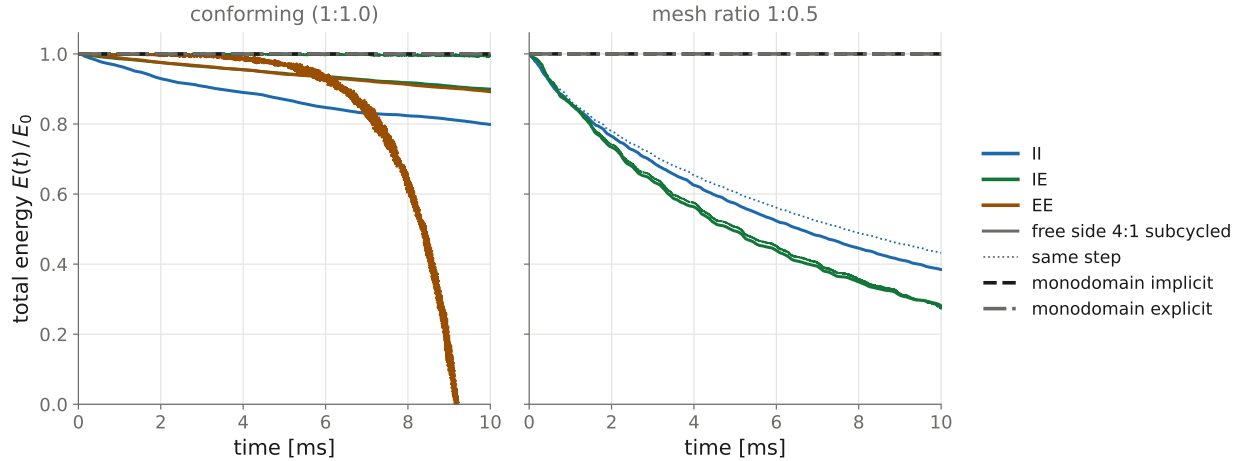


Figure 4: Subcycled (free side 4:1, solid) vs. same-step (dotted) adjoint-paired nonoverlap coupling over 10 ms. Left, conforming: the subcycled EE run (orange solid, CFL = 0.25) avoids the saturation that destroys same-step EE (orange dotted), while subcycled II (blue solid) pays the compounding interpolation dissipation that same-step II (on the reference line) does not. Right, mesh ratio 1:0.5: at strong contrast the trace-jump dissipation dominates and the subcycling overhead is comparatively small (II subcycled -61.6% vs. same-step -56.8%).

The long-horizon ranking is unambiguous: adjoint-paired nonoverlap coupling with at least one implicit interface member, on conforming or nested mesh ratios, is the only tested configuration that is sound at 10 ms; the overlap family is short-horizon only.

12 Windowed controller stops and the relaxation state

The multi-domain controller advances in *stops*; within a stop each subdomain subcycles to the stop end with its own time step, and partner data are exchanged as per-substep trajectories with piecewise-linear time interpolation (Sec. 10). With the subdomain steps held fixed ($1\ \mu\text{s}$), does the controller stop size — the window over which the Schwarz iteration converges before committing — affect the converged physics? A sweep over windows of 1 (baseline), 2, 5, and $10\ \mu\text{s}$ on representative overlap and nonoverlap cases, run to 10 ms, gave a split answer.

Overlap: a pure cost parameter. Every overlap trajectory is identical across all four windows to all printed digits — the conforming II run ends at $+0.61\%$ in all four, and the 1:0.5 instability even dies at the same instant. The overlap exchange interpolates partner kinematics from history with no per-pair relaxation state, so the windowed waveform iteration converges to the same-step fixed point.

Nonoverlap: the window shifted the fixed point. Drift for the conforming paired II case at 10 ms: $+0.00\%$ same-step, $+11.3\%$ at the $2\ \mu\text{s}$ window, -81.9% at $5\ \mu\text{s}$, -83.7% at $10\ \mu\text{s}$ (Aitken relaxation; with fixed $\theta = 0.5$ the sequence is a monotone drain instead, -7.1% at $2\ \mu\text{s}$ to -87.9% at $10\ \mu\text{s}$). This was not a convergence failure — the sweeps converged normally — but a defect of the relaxation machinery: the relaxation state (λ vectors and Aitken residual buffers) was stored as one vector per interface pair, updated on every boundary-condition application. In a windowed stop the relaxed side of a pair (the later-listed member) applies its coupling condition once per *substep*, so the blend $(1 - \theta)g + \theta \text{rhs}$ mixed iterates across *time* rather than across Schwarz sweeps: a causal exponential low-pass filter (lag $\sim \Delta t_{\text{sub}}/\theta$) on the exchanged traction, whose lag dissipation grows with the window. Under Aitken the defect changes sign: the θ estimate compares residuals belonging to different times, overshoots past 1, and the resulting phase lead injects energy — the $+11.3\%$ entry. (The subcycled study of Sec. 10 escaped by an ordering accident: its subcycling free side is listed first, i.e. unrelaxed.)

Two experiments isolate and confirm the mechanism. First, $\theta = 1$ — which uses no relaxation state at all — reproduces the same-step physics on the unmodified code at every window ($+0.00\%$ over the full 10 ms at the $10\ \mu\text{s}$ window). Second, with the fix below in place, all windowed policies land on the same-step trajectory. The fix keys the relaxation state by pair *and* by substep time slot within the current stop, clearing it at every stop (`relaxation_slot!` in `schwarz.jl`); with one substep per stop it reduces exactly to the classical per-pair iteration, and the same-step code path is bit-identical to the pre-fix baselines. Windowed runs then match same-step to Schwarz truncation (the residual 3.9×10^{-6} displacement mismatch at working tolerances collapses to 3.7×10^{-8} as the stopping tolerances tighten), and a windowed-equivalence regression guards this in the adjoint-pairing test (test 109).

Relaxation policy inside windows. With the state fixed, the remaining question is which relaxation policy a windowed stop should use. Measured on the conforming II case at the $10\ \mu\text{s}$ window (mean Schwarz sweeps per stop): $\theta = 1$ converges in 19.1 and runs the full 10 ms cleanly; per-slot Aitken takes ~ 25 and dies at 3.9 ms when an unclamped θ excursion feeds the Newton solver a divergent iterate; fixed $\theta = 0.5$ takes 47.5 (clean); and a waveform-pooled Aitken (one θ from residuals pooled across the window’s slots) takes 55.1 — clean on II but fatal on EE, where it locks θ persistently negative. Aitken acceleration is therefore at best useless and at worst fatal inside windows, and Norma applies it only to single-slot stops, where its $\approx 3\times$ sweep reduction is

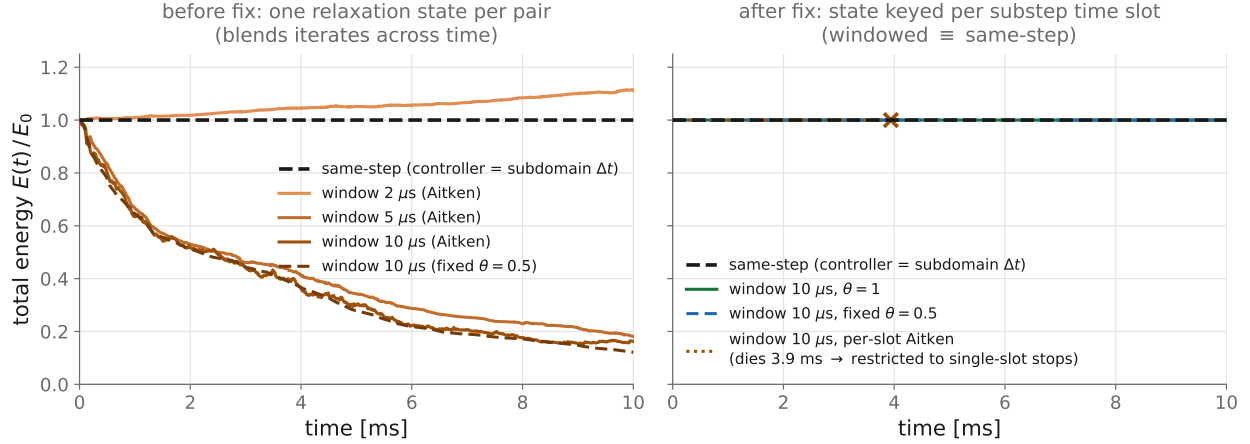


Figure 5: Windowed controller stops on the conforming paired II case, 10 ms. Left, pre-fix: the per-pair relaxation state blends iterates across time — Aitken injects at the $2\mu\text{s}$ window and drains at $5\text{--}10\mu\text{s}$; fixed $\theta = 0.5$ drains monotonically. Right, post-fix (slot-keyed state): every windowed policy lands on the same-step trajectory; the per-slot Aitken run is physically correct until its unclamped θ excursions cause a Newton failure at 3.9 ms ($\times$), which motivated restricting Aitken to single-slot stops.

proven (`aitken_applies`); windowed stops use the configured `relaxation parameter`, with $\theta = 1$ recommended — the impedance dashpot supplies the contraction, so no under-relaxation is needed.

Two limitations round out the picture. The all-explicit pair at large windows is impractical under any policy: the windowed explicit sweep map is nearly non-contractive (fixed $\theta = 0.5$ reaches the 128-sweep limit and still dies at 9.8 ms). And windows are never a cost optimization for the nonoverlap exchange: the number of substep solves per unit time is (sweeps per stop)/ Δt_{sub} regardless of the window, and the windowed sweep counts above all exceed the same-step Aitken count, so the controller step is purely a convenience parameter (e.g. for aligning output stops), cost-neutral at best. For the overlap family, by contrast, larger stops are a mild speedup (fewer, marginally more iterated stops). One earlier observation is withdrawn as an artifact: the pre-fix “EE at the $2\mu\text{s}$ window with fixed θ ends at -0.25% ” entry was the defect’s low-pass filter accidentally damping the grid-frequency saturation; post-fix, windowed EE saturates exactly like same-step EE, and the honest remedy remains CFL margin via subcycling (Sec. 11).

Open issue: a θ -sensitive fixed point. The windowed-equivalence test exposed a pre-existing defect, orthogonal to the fix: the *converged* paired-impedance solution depends weakly on the relaxation parameter — $\theta = 0.5$ and $\theta = 1.0$ differ by $\sim 4 \times 10^{-5}$ relative displacement on the 2:1 nonconforming cantilever at a 10^{-13} stopping tolerance, same-step and windowed alike, and an Aitken run lands on the $\theta = 1$ answer to 10^{-12} . Instrumentation rules out the blend algebra (the blended base is identically zero at interface degrees of freedom, because `apply_bcs` zeroes the boundary force before each pass); the evidence points to a neutral — non-contracting — mode of the paired exchange that retains its θ -dependent initialization and is invisible to the displacement-based stopping criterion. The magnitude is far below every signal in this study ($\geq 10^{-4}$), but it is a genuine correctness question and needs its own investigation.

13 Conclusions and recommendations

13.1 Summary of findings

1. On a shared interface, Robin coupling is fully reflecting ($|R| \equiv 1$) and the impedance condition is absorbing ($|R| < 1$, zero at $\alpha = 0$) with the impedance fixed by material data; in the quasi-static limit the impedance condition collapses onto Robin (Sec. 2).
2. On an overlap boundary, the impedance condition is consistent with the monodomain solution only with the partner-traction term, and the quality of that traction sets the conforming energy error: $59\% \rightarrow 14\% \rightarrow 1.5\% \rightarrow 0.08\%$ over 1 ms as the traction evaluation improves to the consistent nodal reaction, with no adjustable parameters (Sec. 4).
3. On nonconforming overlap decompositions the coupling family is absorbing-or-unstable: Dirichlet matching (pointwise or weak) is severely unstable, and the impedance condition is stable precisely because its dashpot continually absorbs the inter-grid disagreement, whose dominant component is coarse-grid dispersion — a smooth, in-band solution defect that neither spatial filtering nor any measured increase in transfer fidelity can close at fixed h (Secs. 5, 6).
4. The interface work of the overlap coupling is sign-indefinite in principle (two independent, non-adjoint transfers) and in practice (injection up to +275%; sign flips between adjacent mesh ratios), and at a 10 ms horizon 74 of 108 nonconforming overlap runs diverge. The nonconforming overlap coupling cannot be trusted with an energy budget (Secs. 6.4, 11).
5. Energy consistency requires a single-valued interface traction acting through L^2 -adjoint transfer operators built from one cross-mass matrix — then the interface power telescopes to a sign-definite dissipation on the interface jump (14). This is implementable only on the nonoverlap variant, which has a shared interface (Sec. 7).
6. The adjoint-paired nonoverlap coupling — d’Alembert interface reaction, shared cross-mass matrix, harmonic-mean pair impedance, IMEX interface rows for explicit subdomains — never injects at any mesh ratio in any integrator combination, conserves to $\leq 0.3\%$ at 1 ms on conforming and mildly nonconforming meshes in all four combinations, and reproduces the monodomain trajectory pointwise (Sec. 9).
7. At long horizons the paired coupling’s sign-definite dissipation at strong mesh contrast compounds (40–85% at 10 ms); the non-nested ratios 1:0.7–1:0.8 accumulate interface-local high-frequency content until an interface element inverts; and the all-explicit pair at CFL = 1 saturates with undamped grid-frequency content, for which subcycling (CFL margin) is an effective remedy (Sec. 11).
8. Subcycling uses genuine piecewise-linear interpolation of partner trajectories (two silent defects that reduced it to end-sampling are fixed); its $O(\Delta t)$ dissipation is the price of subcycling, and exact interface-work telescoping is structurally incompatible with a partitioned exchange — it requires a direct interface solve (Sec. 10).
9. The Schwarz relaxation state must be keyed per pair and per substep time slot; otherwise controller stops larger than a subdomain step blend iterates across time and shift the converged fixed point. With the slot-keyed state, windowed stops are equivalent to same-step to Schwarz truncation, and the controller step is a pure convenience parameter. Aitken acceleration applies only to single-slot stops; windowed stops should use $\theta = 1$ (Sec. 12).

13.2 Practical recommendations

- **Energy-critical dynamic coupling:** use the adjoint-paired nonoverlap impedance variant (the default for `Schwarz impedance nonoverlap`) with at least one implicit interface member and a conforming or nested mesh ratio. Avoid the non-nested mild ratios (1:0.7–1:0.8 class) for long runs.
- **Overlap decompositions:** conforming meshes with implicit–implicit integration conserve energy at any horizon; every other overlap configuration is a short-transient method. On conforming node-aligned interfaces the consistent traction makes the impedance overlap condition exact to the Dirichlet-coupling reference.
- **All-explicit interfaces:** provide CFL margin (e.g. subcycle the finer side); central difference at CFL = 1 cannot remove the grid-frequency content the exchange seeds.
- **Parameters:** no impedance tuning anywhere — the impedance is material data (harmonic mean per pair for nonoverlap; partner material with P/S split for overlap). The Robin parameter affects only the iteration, not the converged solution, and must be a single value per interface under pairing. Historical per-problem tuning of impedance scales and large Robin values is obsolete.
- **Diagnostics:** use the blended (partition-of-unity) energy output and the staggered metric (3) for explicit members; use the channel-resolved interface-work instrumentation as the acceptance metric for any change to the transmission machinery — a small net drift can be a cancellation of large opposing channels.
- **Robustness at fixed cost** on nonconforming overlap runs that cannot be avoided: mild HHT- α damping ($\bar{\alpha} \approx 0.02$ – 0.05) converts sign-indefinite drift into monotone decay, at a cost proportional to the energy in the reverberating band.

13.3 Open questions

(i) The weakly θ -sensitive fixed point of the paired exchange ($\sim 4 \times 10^{-5}$ relative displacement), evidence of a neutral mode invisible to the displacement-based stopping criterion (Sec. 12). (ii) The non-nested mesh-ratio failure window of the paired transfer at long horizons (Sec. 11). (iii) Exact subcycled telescoping via a direct FETI-style interface solve, together with the localized-Lagrange-multiplier frame construction (Park–Felippa; Dvorak et al. 2023) (Sec. 10.2). (iv) A bimaterial benchmark to exercise the tensor impedance and the harmonic-mean pair impedance where they should be most beneficial. (v) A smooth-content benchmark to quantify the HHT- α robustness/cost trade on problems with scale separation (Sec. 5.3).

Data provenance

Cantilever ladder study and instrumentation: `rigel:~/test/Norma/cantilever-energy`. Parametric study, ratio sweep, 10ms horizon, controller-step sweep, figures, and run harness: `~/cantilever-energy-study/ (ratio-sweep/, *-10ms, controller-step-sweep/, wccm-plots/, drift-table-10ms.txt)`. Regression tests: 103/104 (overlap projectors and consistent traction), 105 (content-absorption dissipativity), 109 (adjoint pairing and windowed equivalence, `test/schwarz-nonoverlap-impedance-adjoint-pairing.jl`).